

# KNN → K-Nearest Neighbours

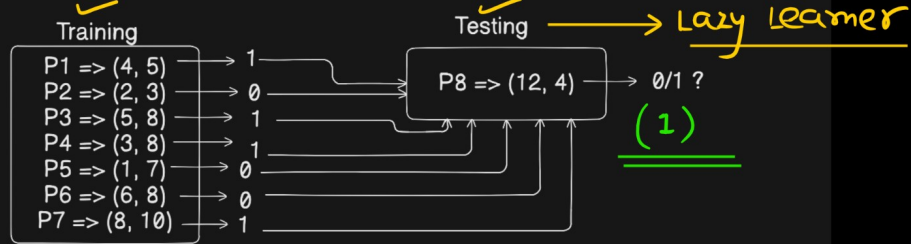
don't prefer to Distance evaluation → Euclidean

very  
high  
amount  
of data  
(scalable)

use KNN  
k = 3

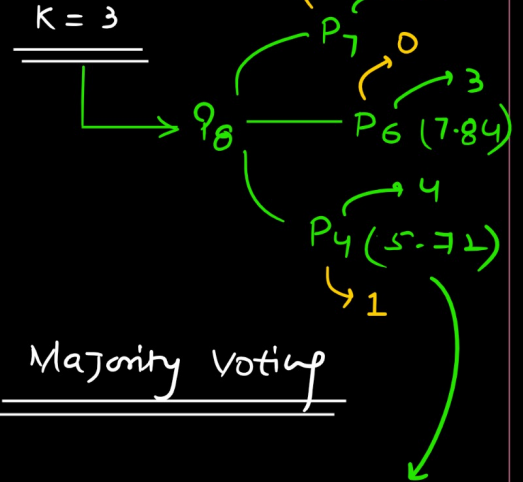
Classification & Regression

Distance



Nearest  
neighbours

$$\begin{aligned} P_1 - P_8 &\Rightarrow 5 \\ P_2 - P_8 &\Rightarrow 7 \\ P_3 - P_8 &\Rightarrow 6 \\ P_4 - P_8 &\Rightarrow 4 \leftarrow \\ P_5 - P_8 &\Rightarrow 10 \\ P_6 - P_8 &\Rightarrow 3 \leftarrow \\ P_7 - P_8 &\Rightarrow 2 \leftarrow \end{aligned}$$



Majority Voting

Average

Distance Metric → Euclidean

① Distance

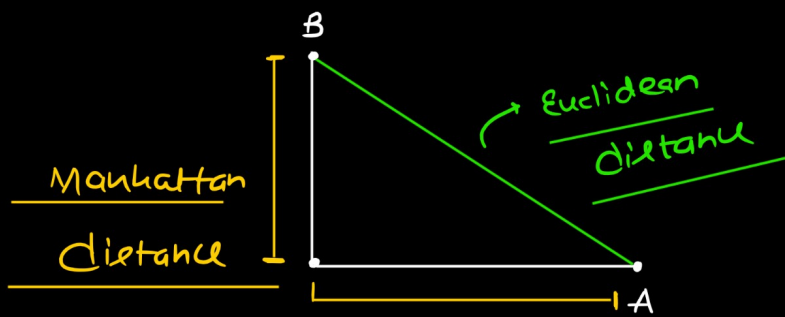
$$\begin{matrix} P_1 & & P_2 \\ (x_1, y_1) & & (x_2, y_2) \end{matrix}$$

$$\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$$

numeric  
type  
of  
data

② Manhattan distance

$$\begin{matrix} (x_1, y_1) & & (x_2, y_2) \\ P_1 & & P_2 \end{matrix} \Rightarrow |y_2 - y_1| + |x_2 - x_1|$$



③ Hamming distance

↑ str1  
↓ str2

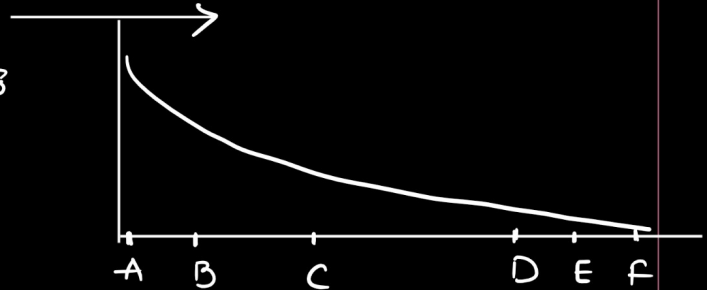
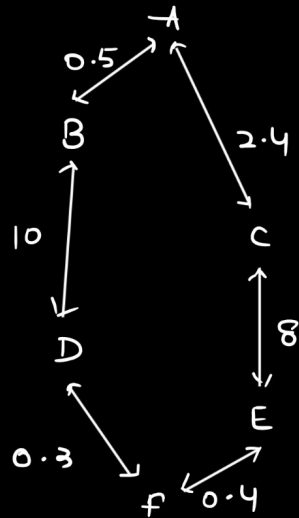
a a b b c d a  
b a b c b d a

3 mismatch b/w two

vectors/  
chars in

a given

string

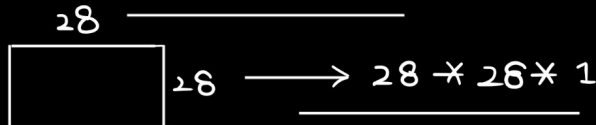


type of image

Grayscale

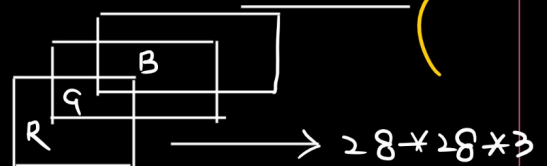
image

1 channel



RGB

3 channels



Pixel

|   |   |
|---|---|
| 7 | 3 |
| 9 | 8 |
| 4 | 5 |

| KNN                              | UMAP                                             |
|----------------------------------|--------------------------------------------------|
| use neighbors for prediction     | use neighbors for dimensionality reduction       |
| Output $\rightarrow$ class label | output $\rightarrow$ lower dimensional embedding |
| majority voting                  | graph construction                               |